



National University of Computer & Emerging Sciences, Karachi

Fall-2023 Department of Computer Science

Final Exam

26th December 2023, 09:30 AM – 12:00 PM

Part (B)



Course Code: CS2009	Course Name: Design and Analysis of Algorithm
Instructor Name / Names: Dr. Muhammad Atif Tahir, Dr. Fahad Sherwani, Dr. Farrukh Salim, Dr. Nasir Uddin, Mr. Faisal Ahmed, Mr. Sandesh Kumar, Ms. Ansum Hamid, Mr. Syed Faisal Ali, and Mr. Minhal Raza	
Student Roll No:	Section:

Instructions:

- Return the question paper.
- Read each question completely before answering it. There are 8 questions on 4 pages (Q3-Q10).
- In case of any ambiguity, you may make assumptions. But your assumption should not contradict any statement in the question paper.

Time: 150 minutes

Max Marks: 40

Question # 3 [CLO:1] Answer the following questions and explain in only 2-3 lines only. [5 marks]

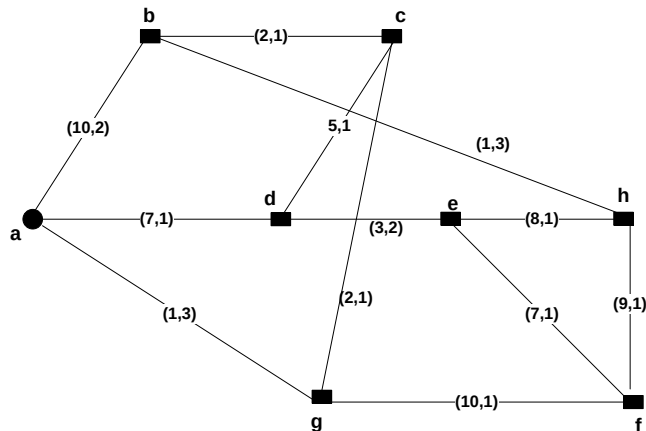
- [0.5 Point] Floyd-Warshall algorithm computes shortest path from every vertex to every other vertex using $O(n^3)$ complexity, whereas Bellman-Ford can compute in $O(n^3)$, only from a single source to other vertices. Why one needs to use Bellman-Ford algorithm, in presence of more information rich Floyd-Warshall algorithm. (apart from detecting a negative cycle)
- [0.5 Points] True / False: Does Directed Acyclic Graph (DAG) can have negative weight edges?
- [0.5 Point] Describe the difference between tight and loose bounds in Big-O notation. Provide an example scenario where a tight bound is preferable and explain why.
- [0.5 Point] What do you mean by the optimal solution?
- [1 Point] Problem A reduces to problem B if you can use an algorithm that solves B to help solve A. Let's say problem A is to find median and problem B uses any sorting method with $O(n^2)$ complexity. What would be the complexity of Problem A
- [0.5 Point] True or False: In dynamic programming, a subproblem do share sub-subproblems
- [1.5 Points] Differentiate b/w Single-source, Single-pair and All-pairs shortest path and one algorithm for each studied during this course

Question # 4 [CLO: 3]

[5 marks]

Demand for multimedia, combining audio, video, and data streams over a network, is rapidly increasing. Some of the most popular uses of multimedia are real-time interactive applications such as desktop video and audio conferencing, collaborative engineering, shared white boards,

transmission of university lectures to a remote audience, and animated simulations. With the advent of real-time interactive applications, **delay constraint** is also an important objective along with minimizing bandwidth cost. Delay constraint means that a packet must be reached within that interval. Figure below shows an example tree. Each edge consists of 2 values (**first one is bandwidth cost** and **second one is delay**). Your objective is to design a spanning tree that optimizes both bandwidth cost and delay constraint simultaneously. Later show the dry run using Figure below. [Hint: Modify Prim's or Kruskal using some weighted cost function].



Question # 5 [CLO: 3]

[5 marks]

Longest palindrome subsequence: A palindrome is a nonempty string over some alphabet that reads the same forward and backward. Examples of palindromes are all strings of length 1, civic, racecar etc. Give an efficient algorithm to find the longest palindrome that is a subsequence of a given input string. For example, given the input “character”, your algorithm should return “carac”. The idea is to check and compare the first and last characters of the string. There are only two possibilities for the same: If the first and the last characters are the same, it can be ensured that both the characters can be considered in the final palindrome and hence, add 2 to the result, since we have found a sequence of length 2 and recurse the remaining substring $S[i + 1, j - 1]$. In case, the first and the last characters aren't the same, the following operation must be performed:

Recurse $S[i + 1, j]$

Recurse $S[i, j - 1]$

- Design an algorithm using a brute force approach (2^n possible subsequences) to find the maximum length obtained amongst them [2 Points]
- Design an algorithm using dynamic programming approach (polynomial time solution required) to find the maximum length obtained amongst them [3 Points]

For both parts (a) and (b), only pseudocode is required but must be with clear formulas / loops/ if-else statements not just plain text. You can show dry run for your own understanding, but dry run would not have any weightage.

Question # 6 [CLO: 1]**[5 marks]**

Given an array of integers `arr[]` of size `N` and an integer, the task is to rotate the array elements to the left by `d` positions, (where $d \leq n$).

a) Design an algorithm to solve the above task with $O(n)$ time complexity, and $O(n)$ space complexity [2.5 Points].

b) Design an algorithm to solve the above task with $O(n^2)$ time complexity, and $O(1)$ space complexity [2.5 Points]

Pseudocode in plain text is acceptable for this question but must be clear.

Example:

Input: `arr[] = {3, 4, 5, 6, 7, 1, 2}`, `d=2`

Output: 5 6 7 1 2 3 4

Question # 7 [CLO:3]**[5 marks]**

Calculate the Levenshtein-Distance for `str1 = "PLASMA"`, and `str2 = "ALTRUISM"` using Dynamic Programming Approach of Levenshtein-Distance.

Pseudocode Levenshtein-Distance is given below.

For two strings `X` of length `n` `Y` of length `m`. We define $D(i,j)$ the edit distance between `X[1..i]` and `Y[1..j]` i.e., the first `i` characters of `X` and the first `j` characters of `Y`. The edit distance between `X` and `Y` is thus $D(n,m)$.

Algorithm

//Initialization $D(i,0) = i$

$D(0,j) = j$

//Recurrence Relation:

For each $i = 1 \dots n$

For each $j = 1 \dots m$

$D(i,j) = \min($

$D(i-1, j) + 1,$

$D(i, j-1) + 1,$

$D(i-1, j-1) + 2, \text{ if } X(i) \neq Y(j)$

$D(i-1, j-1) + 0 \text{ if } X(i) = Y(j))$

//Termination: $D(n,m)$ is distance

Question # 8 [CLO: 5]**[5 marks]**

While visiting the Northern Areas of Pakistan you asked your travel guide to provide you information regarding the famous tourist points. Your travel guide handed you the following table showing four tourist points and the distances between these locations. You are determined to visit

all tourist points once and return to the starting location besides reducing your travel expenses. Based on your knowledge in this course how would travel to these locations. Show all the computations and steps to solve it

	A_1	A_2	A_3	A_4
A_1	0	2	1	4
A_2	2	0	3	2
A_3	1	3	0	5
A_4	4	2	5	0

Question # 9 [CLO: 5]

[5 marks]

Prove that $\phi \in 3SAT \iff G\phi \in VertexCover$, where $\phi = (X_1 \vee X_2 \vee X_3) \wedge (\neg X_1 \vee \neg X_2 \vee \neg X_3) \wedge (\neg X_1 \vee X_2 \vee X_3)$. Let ϕ be a 3-CNF formula with m variables and l clauses. Construct in polynomial-time, an instance of (G, k) where $k = m + 2l$

Question # 10 [CLO: 3]

[5 marks]

Apply the Bellman-Ford shortest path algorithm on the following graph, starting from vertex "a". Show all steps

